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Comparing Real Space and Conventional Method in abTEM

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Table 1: List of abbreviations

Symbol	Description
TEM	Transmission Electron Microscope
λ	Wavelength
Å	Angstrom
$\Psi(\vec{x})$	Wave function
$v_z(\vec{x})$	Projected atomic potential
σ_e	Interaction parameter
$t(\vec{x})$	Transmission function
m	Relativistic mass of electron
m_0	Rest mass of electron
c	Light speed
eV	Electrostatic potential energy of the electrons
V	Accelerating voltage
h	Planck's constant
FT	Fourier Transformation
FT^{-1}	Inverse Fourier Transformation
$g(\vec{x})$	Image intensity
Δz	Distance between slices
$\oplus$	Convolution
ASE	Atomic Simulation Environment

Abstract

Transmission Electron Microscopy is a technique, that enables the observation of very fine details on a specimen by transmitting a beam of electrons through it. Significantly achieving resolution that exceeds conventional light microscopes. Confidently simulating a Transmission Eelectron Microscope enables a much cheaper way of analyzing their images. But trying to simulate a Transmission Electron Microscope is not an easy task and there are different ways to achieve the desired goal. In this bachelor thesis the conventional- and real space multislice method were compared by using the open source packet abTEM, developed by Dr. Jacob Madsen in the research group of Prof. Toma Susi. Different slice thicknesses, sampling sizes and energies were used as parameters to enable a comparison between the two methods.

1 Introduction

In the following bachelor thesis the conventional- and real space multislice method for simulating transmission electron microscopy will be compared by using a set of different slice thicknesses, sampling sizes and energies. The results can be seen in section 3 and will be discussed in section 4. For that purpose the open source package abTEM (developed by Dr. Jacob Madsen) will be used to calculate the necessary data, as can be seen in section 2. Using Jupyter Notebook (written in Python), allows a quick understanding and reproducibility of the used code and documentations by allowing the implementations of subtasks.

The aim of this bachelor thesis is to compare the conventional- and real space multislice method and to gather enough data by using different sets of slice thicknesses, sampling sizes and energies to be able to conclude any meaningful difference between the two methods, corresponding to accuracy and computational time. What the advantages of the real space method over the conventional method are and if there is even a difference will be concluded at section 4.

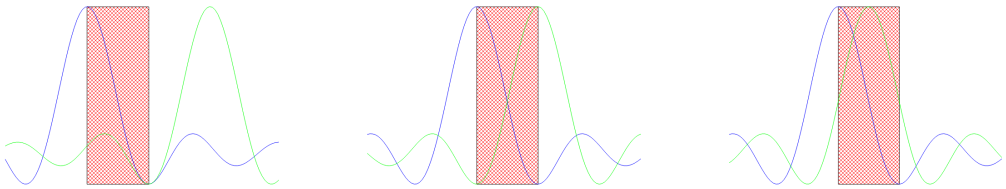
Before starting with the computation, a quick overview over relevant theory behind resolution 1.2, the structure of Transmission Electron Microscopes 1.3 and the theory of simulating TEM 1.4 will be given.

1.1 Transmission Electron Microscope

A Transmission Electron Microscope (TEM) enables the observer to see very fine details, much finer than that would be possible with just the human eye or even a classical light microscope. By using electrons instead of light waves, a TEM achieves a much higher resolution, down to a few nanometers.

1.2 Resolution

Resolution plays a big part in Transmission Electron Microscopy, but how can resolution be defined? By following John William Strutt, 3rd Baron Rayleigh. Resolution is the limit at which an observer can still distinguish two points. Concluding that two point sources can still be distinguished if the minimum of one source is under the maximum of the other [WC09, P. 107]. This is called the Rayleigh-Criteria and the three different outcomes can be seen in Figure 1.



(a) (b) (c)

Figure 1a shows the two point sources far enough away so both can be easily seen separately.
Figure 1b shows the minimum distance between two point sources. Both can still be distinguished.
Figure 1c shows the two point sources too close to each other. Both can no longer be seen separately.

Figure 1: Visualized Rayleigh-Criteria

The Rayleigh-Criteria is not a fundamental physical rule but nonetheless a practical definition to determine the resolution [WC09, P.107].

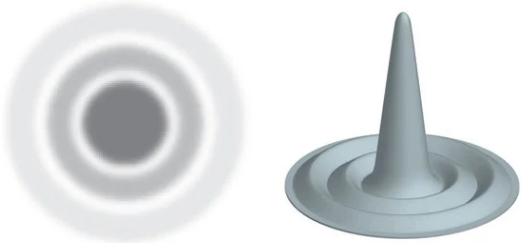


Figure 2: Airy disk [MW23]

Having established the Rayleigh-Criteria. The next question would be how close can two point sources be? Picturing a point source from the top as can be seen in Figure 2, each point source has a central Airy disk diameter which can be determined by Equation 1. Where λ is the wavelength, b is the distance from the lens to the image layer and D the diameter of the lens [Dem12, P.358].

$$d_{airy\ disk} = 2,44 \cdot \lambda \cdot b/D \tag{1}$$

The Rayleigh criterion states that two point sources can still be resolved if the minimum of one source is below the maximum of the other. Therefore, the distance between the two maxima must be at least half the diameter of the central airy disc [Dem12, P.358].

$$\Delta x_{min} = 1/2 \cdot d_{airy\ disk} = 1,22 \cdot \lambda \cdot b/D \tag{2}$$

The maximum aperture angel captured by the objective lens is defined by $2sin(\alpha) = D/f$. In the general case the objective plane lies in the focal plane of the objective lens. Resulting in $f \approx g$ [Dem12, P.358].

$$\Delta x_{min} = 1,22 \frac{\lambda}{2sin(\alpha)} \tag{3}$$

The change in the wavelength between the objective and the lens can be described by $\lambda_n = \lambda_0/n$, where n is the refractive index [Dem12, P.358].

$$\Delta x_{min} = 0,61 \frac{\lambda_0}{nsin(\alpha)} = 0,61 \frac{\lambda}{NA} \tag{4}$$

By using an immersion oil with a refractive index of $n = 1.5$ and with an maximum aperture angel of $2\alpha = 106°$ the distance where two point sources can still be resolved is limited by half the wavelength [Dem12, P.359].

$$\Delta x_{min} = 0,5\lambda \tag{5}$$

For example with the same setup as in Equation 5, green light with a wavelength of around 550 nm, enables the distinguishment of two points with a distance up to 275 nm. This is theoretically the best case. Practically the resolution for a good light microscope, considering green light, lies at around 300 nm [WC09, P. 5]. To achieve better resolution smaller wavelengths need to be used. Here come electrons into play.

The wavelike characteristics of electrons were first theorized by Louis de Broglie in 1925 and two years later experimentally proven by two independent research groups. In 1932 Knoll and Ruska first mentioned the idea of an microscope using electrons. And just four years later the first TEM was build [WC09, P. 4].

The idea for an electron microscope comes from the wavelike characteristic of electrons. Louis the Broglie's equation 6 shows that the wavelength of electrons is related to their energy [WC09, P. 6].

$$\lambda = 1.22 \cdot E^{-1/2} \tag{6}$$

Theoretically for an energy of 100 keV, equation 6 would lead to a wavelength of approximately 0.004 nm. Much much smaller than the light waves previously mentioned. Of course such a resolution will be practically not be feasible. A perfect electron lens does not exist right now, leaving most TEMs with an resolution of < 1 nm to 100 nm. With a few TEMs reaching resolution down to 0.1 nm. [WC09, P.5-6].

1.3 The Instrument

The resolution of a good optical light microscope lies at around 300 nm, while a Transmission Electron Microscope easily achieves images with a resolution of around 1 - 100 nm by taking advantage of the small wavelength of electrons. With the improvement of spherical aberration correction in objective lenses, even resolution below 1 Å can be achieved [WC09, P.5-6].

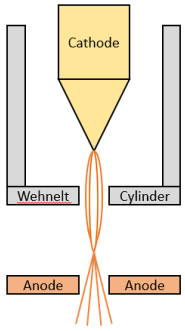
To accomplish such a feat, electrons are first generated in an electron gun. There are two main types of electron guns as seen in Figure 3a & 3b.

A thermionic gun works by resistively heating a metal wire to cause thermionic emission. Leading to the emitted electrons to be attracted and accelerated by the opposing anode. A Wehnelt cylinder laying between the cathode (metal wire) and anode functions as a simple electrostatic lens, able to control the electron beam [WC09, P.77].

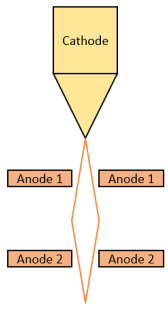
The second type is a Field-Emission Gun (short FEG) consisting of one cathode and two anodes. The cathode is shaped in a way that it ends in a very small tip, easing the extraction of electrons. First the opposing anode is charged by several kV building up the extraction voltage needed to enable the electrons in the cathode tip, to tunnel out of it. The second anode then attracts and accelerates the electrons. Whereby the field of the anodes act like an electrostatic lens able to limitly control the electron beam [WC09, P.80]

After the electron gun, the beam then passes through the condenser lenses. Magnetic lenses able to control the electron beam in three degrees of freedom (angle, current, focus). Allowing the transportation of the electron beam onto the specimen [Kir20, P.12].

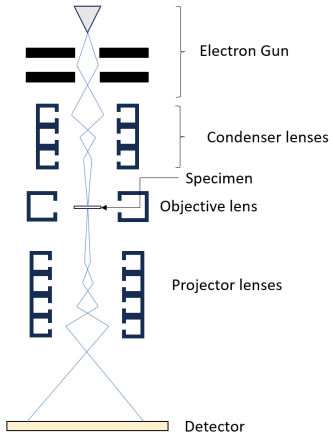
The next step happens when the electron beam enters the objective lens and is getting focused onto the specimen. To achieve a good image the diameter of the electron beam should not be larger than 5 nm[WC09, P.8].
The specimen itself requires to be very thin and should not exceed a thickness of 100 nm. Keeping in mind the saying "thinner is always better" can be applied in the case of the specimen. If the specimen is thin enough, sufficient electrons are able to transmit through the specimen allowing enough intensity to fall onto the detector below[WC09, P.11].
The projector lenses further magnify the image [Kir20, P.12].



(a) Termionic Gun



(b) Field Emission Gun



(c) Transmission Electron Microscope

Figure 3: Simple Illustration of the two electron guns and TEM

1.4 Simulating

By simulating a Transmission Electron Microscope a lot of the previously mentioned parts in chapter 1.3 can be ignored. For example it is not necessary to simulate the generation of the electrons with an electron gun or the following control of the electron beam with the condenser lenses. One can immediately start with a perfectly aligned electron beam ready to transmit through the prepared specimen. The projector lenses can also be ignored, because any defects in the objective lens are greatly magnified and the angles in the projector lenses are greatly reduced leaving little effect on the final image. Requiring, for the most part only the objective lens and the specimen to be simulated. Shrinking the before seen TEM-structure in Figure 3 to what can be now seen in Figure 4 [Kir20, P.13]

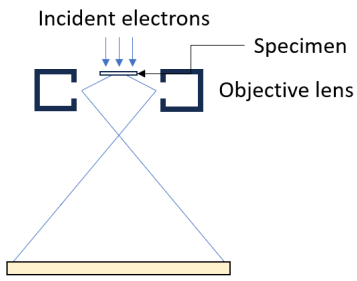


Figure 4: Simplified illustration of an TEM showing only the parts needed for simulating

1.4.1 Image simulation for thin specimens

A thin specimen can be interpreted as a single slice in the x-y-axis, with the electron beam traveling down the z-axis, as can be seen in Figure 5 [Kir20, P.99-100].

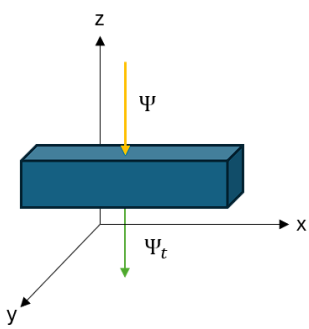


Figure 5: Illustration of the incident wave function Ψ entering a thin specimen and leaving as the transmitted wave function Ψ_t

The electron beam itself can be understood as a plane wave, with its wave function $\Psi(\vec{x})$ being the incident wave function 7 [Kir20, P.100].

$$\Psi(\vec{x}) = exp(2\pi i k_z z) \tag{7}$$

With $k_z = 1/\lambda$ being the propagation wave vector and λ the wavelength.
In general the electrons that are transmitted through the specimen have a far higher energy then the electrons in the specimen. Resulting only in a small deviation in their path, which can be approximated as a small change in wavelength λ. This change in wavelength can also be interpreted as a shift in phase [Kir20, P.100].

The weak phase object approximation assumes for very thin specimens that the shift in phase can be seen as a specimen transmission function $t(\vec{x}) = exp(i\sigma_e v_z(\vec{x}))$. Therefor the transmitted wave function $\Psi_t(\vec{x})$ is just the incident wave function $\Psi(\vec{x})$ multiplied by the specimen transmission function $t(\vec{x})$ [Kir20, P.100-102].

$$\Psi_t(\vec{x}) = t(\vec{x}) \cdot \Psi(\vec{x}) = exp(i\sigma_e v_z(\vec{x})) \cdot exp(2\pi i k_z z) \tag{8}$$

With $v_z(\vec{x})$ being the projected atomic potential and σ_e the interaction parameter.

To be able to simulate a TEM image, all positions of the atoms need be to be known. Then, with a linear superposition of the potential for each atom in the specimen (assuming all the atoms are independent), the atomic potential $v_z(\vec{x})$ can be calculated via equation 9 [Kir20, P.111-112].

$$v_z(\vec{x}) = \sum_{j=1}^{N} v_{zj}(\vec{x} - \vec{x}_j) \tag{9}$$

where $\vec{x}_j$ is the position of atom j and v_{zj} the projected atomic potential.
Equation 9 is the potential for the whole specimen and can be used to calculate the transmission function $t(x)$ with the interaction parameter σ_e being [Kir20, P.101,111]:

$$\sigma_e = \frac{2\pi}{\lambda V}\left(\frac{m_0 c^2 + eV}{2m_0 c^2 + eV}\right) = \frac{2\pi m e \lambda}{h^2} \tag{10}$$

With m_0 being the rest mass of the electron, $m = \gamma m_0$ the relativistic mass, c the light speed, eV the electrostatic potential energy of the electrons, V the accelerating voltage and h Planck's constant.

The incident wave function $\Psi(\vec{x})$ from equation 8 can be ignored, because in the end only the intensity matters which is unaffected by the incident wave function. The transmitted wave function $\Psi_t(\vec{x})$ is therefor just the transmission function $t(\vec{x})$ [Kir20, P.114].

$$t(\vec{x}) = exp(i\sigma_e v_z(\vec{x})) = \Psi_t(\vec{x}) \tag{11}$$

A TEM image is cylindrical symmetric and it is therefor essential that the simulation preserves that symmetry. By bandwidth limiting both the projected atomic potential $v_z(\vec{x})$ and the transmission function $t(\vec{x})$ the appearance of various artifacts can be avoided [Kir20, P.114-115].

After the electron beam has transmitted through the specimen. The aberrations of the objective lens can be simulated by Fourier transforming the transmitted wave function $\Psi_t(\vec{x})$ and then multiplying it by the transfer function $H_0(k)$ of the objective lens [Kir20, P.115].

$$\Psi_j(\vec{k}) = FT[\Psi_t(\vec{x})] \cdot H_0(k) \tag{12}$$

The last steps to achieve the final image are then to Fourier transform $\Psi_j(k)$ back to real space $\Psi_j(\vec{x}) = FT^{-1}[\Psi_j(\vec{k})]$ and to calculate its square modulus to get the final image intensity $g(\vec{x})$ [Kir20, P.115-116].

$$g(\vec{x}) = |\Psi_j(\vec{x})|^2 \tag{13}$$

A summary of all the steps necessary can be viewed from Table 2 cited out of [Kir20, P.116].

Step 1	Calculate the projected atomic potential $v_z(\vec{x})$.
Step 2	Calculate the transmission function $t(\vec{x})$ and symmetrically bandwidth limit it.
Step 3	The transmitted wave function $\Psi_t(\vec{x})$ is equal to the transmission function $t(\vec{x})$.
Step 4	Fourier transform the transmitted wave function $\Psi_t(\vec{k}) = FT[\Psi_t(\vec{x})]$ and multiply it by the transfer function $H_0(k)$ of the objective lens to get the effects of the abberations of the objective lens $\Psi_j(\vec{k}) = \Psi_t(\vec{k}) \cdot H_0(k)$.
Step 5	Inverse Fourier transform back to real space $\Psi_j(\vec{x}) = FT^{-1}[\Psi_j\vec{k}]$.
Step 6	Square modulus to get the image intensity $g(\vec{x}) = \lvert\Psi_j(\vec{x})\rvert^2$.

Table 2: Steps for simulating TEM images for thin specimens [Kir20, P.116]

1.4.2 The Multislice Method

In thick specimens the electrons interact strongly when passing through the specimen, allowing scattering to happen more than once. Complicating the calculation of the electrons trajectory. Cowley and Moodie derived the conventional multislice method by dividing the specimen into thin two-dimensional slices in which the electron beam gets alternately transmitted through a slice and propagated to the next one [Kir20, P.143-44].

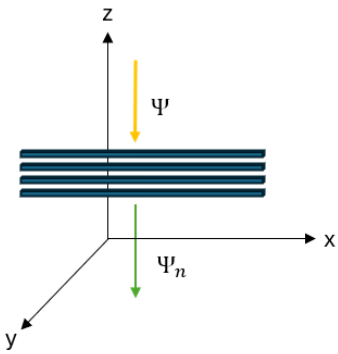


Figure 6: Illustration of the specimen divided into multiple thin slices

The transmitted wave function $\Psi_{n+1}(x, y)$ for each slice can be written as seen in Equation 14 [Kir20, P.165].

$$\Psi_{n+1}(x, y) = p_n(x, y, \Delta z_n) \oplus [t_n(x, y)\Psi_n(x, y)] \tag{14}$$

with $p_n(x, y, \Delta z_n)$ being the propagator function, $t_n(x, y)$ the transmission function and $\oplus$ a two-dimensional convolution. [Kir20, P.165].

$$p_n(x, y, \Delta z_n) = FT^{-1}[exp(-i\pi\lambda k^2\Delta z)] \qquad t_n(x, y) = exp(i\sigma_e v_{zn}(x, y)) \tag{15}$$

$\Psi_n(x, y)$ is the previous wave function and Δz is the distance between the slices. The derivation to get to equation 14 can be looked up in [Kir20, P.162-165].

To be able to calculate the exit wave function $\Psi_n(x, y)$, first the projected atomic potential $v_{zn}(\vec{x})$ from Equation 9 must be calculated for each slice. With $v_{zn}(\vec{x})$, the transmission function $t_n(\vec{x})$ can be calculated for each slice. Now the wave function $\Psi_{n+1}(x, y)$ can be calculated for each slice by initializing equation 14 with $\Psi_0(x, y) = 1$. Resulting in the exit wave function $\Psi_n(x, y)$. The steps happening after exiting the specimen are the same as in section 1.4.1 and are summarized in table 3 cited out of [Kir20, P.181].

Step 1	Divide the specimen into thin slices.
Step 2	Calculate the projected atomic potential $v_z(\vec{x})$ for each slice and symmetrically bandwidth limit it.
Step 3	Calculate the transmission function $t(\vec{x})$ for each slice and symmetrically bandwidth limit it.
Step 4	Initialize the incident wave function $\Psi_0(x, y) = 1$.
Step 5	Recursively transmit and propagate the wave function through each slice $\Psi_{n+1}(x, y) = p_n(x, y, \Delta z_n) \oplus [t_n(x, y)\Psi_n(x, y)]$
Step 6	Fourier transform the exit wave function $\Psi_n(\vec{k}) = FT[\Psi_n(\vec{x})]$ and multiply it by the transfer function $H_0(k)$ of the objective lens to get the effects of the aberrations of the objective lens $\Psi_j(\vec{k}) = \Psi_n(\vec{k}) \cdot H_0(k)$.
Step 7	Inverse Fourier transform back to real space $\Psi_j(\vec{x}) = FT^{-1}[\Psi_j\vec{k}]$.
Step 8	Square modulus to get the image intensity $g(\vec{x}) = \lvert\Psi_j(\vec{x})\rvert^2$.

Table 3: Steps for simulating TEM images for thick specimens - Multislice method [Kir20, P.181]

1.4.3 Real Space Method

Another way to calculate the exit wave is through the real space method introduced by Van Dyck, which in principle takes nearly the same steps as the conventional multislice method, seen in table 3, but calculates the convolution and transmission in real space [Kir20, P.144]. The conventional multislice method uses the Fourier transformation to turn complicated derivatives into multiplication as seen in Equation 16 [Kir20, P.170],[WS15, P.213].

$$\begin{aligned} f(\vec{x}) \oplus h(\vec{x}) &= FT^{-1}[F(\vec{k}) \cdot H(\vec{k})] \\ F(\vec{k}) = FT[f(\vec{x})] \quad &; \quad H(\vec{k}) = FT[h(\vec{x})] \end{aligned} \tag{16}$$

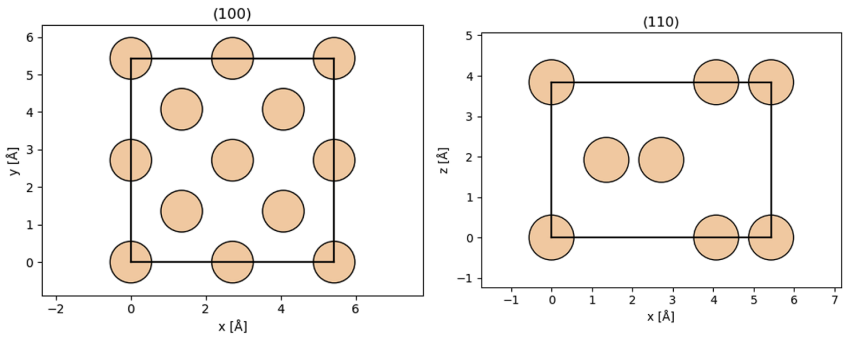
The real space multislice method stays in real space and approximates the derivatives by a finite difference scheme derived from the Taylor expansion. This calculation can diverge for specific parameters as can be later seen in section 2.4 [WS15, P.213-214].

2 Method

To compare the conventional and real space multislice method the open source package abTEM was used. It is mainly written in python and allows the user to compute the exit wave of the electrons passing through the specimen.

2.1 Preparation

In the case of this bachelor thesis a diamond cubic shaped silicon structure was selected and used as the specimen, as can be seen in Figure 7a. ASE (short for Atomic Simulation Environment) was used to build the initial silicon structure. To allow optimal multislicing the structure was rotated by 45° degrees in the x-axis, changing the shape from diamond-cubic to hexagonal (fig. 7b).



(a) Diamond Silicon Structure (b) Hexagonal Silicon Structure

Figure 7: Silicon Structure

With the rotation, the miller plane changed from (100) to (110). The miller plane can be seen as a plane normal to the electron wave. Code 1 was used to get the desired structures seen in Figure 7.

```
rotation = 45
axis = "x"
silicon = ase.build.bulk("Si", cubic = False)

#Rotating silicon structure by 45 degrees.
rotated_silicon = silicon.copy()
rotated_silicon.rotate(rotation, axis, rotate_cell = True)
rotated_silicon, transform = abtem.orthogonalize_cell(
    rotated_silicon, max_repetitions=10, return_transform=True
)

fig, (ax1, ax2) = plt.subplots(1,2, figsize=(10,4))
abtem.show_atoms(
    ase.build.bulk("Si", cubic = True),
    ax=ax1,
    show_periodic = True,
    scale = 0.5,
    title = "(100)"
)
```

```
abtem.show_atoms(
    rotated_silicon,
    ax=ax2,
    show_periodic = True,
    scale = 0.5,
    title="(110)"
)
plt.tight_layout()
```

Code 1: Preperation of the silicon structure

2.2 Parameters

To be able to compare the two multislice methods different slice thicknesses, sampling sizes and energies were used.

For the slice thicknesses, fractions of the cell height a_z ranging from $a_z \approx 3.840\AA$ to $a_z/256 \approx 0.015\AA$ were calculated using code 2.

```
thickness_fractions = np.array([2**n for n in np.arange(0,9)])
cell_height = rotated_silicon.cell[2,2]
slice_thicknesses = cell_height/thickness_fractions
```

Code 2: Slice Thicknesses calculation

The sampling sizes were arranged starting from 0.01 Å and going up to 0.2 Å in 0.01 Å steps using the numpy "arange" function 3.

```
sampling = np.arange(
    start = 0.01,
    stop = 0.21,
    step = 0.01
)
```

Code 3: Sampling size calculation

In the same way the energies, starting from 80 keV and increasing 10 keV every step up to 300 keV, were arranged with code 4.

```
energies = np.arange(
    start = 80e3,
    stop = 310e3,
    step = 10e3,
    dtype = int
)
```

Code 4: Energy size calculation

2.3 Computing the exit wave

With the parameters set, the exit wave can be calculated with a number of steps.

First the potential unit of the specimen will be calculated with the "abtem.Potential()" function. Taking the prepared specimen, slice thickness and sampling size, parametrization and projection as parameters (as seen in code:5). The parametrization (labato or kirkland) describes the radial dependence of the potential for each element and the projection (finite or infinite) describes for finite if the 3D potential is numerically integrated between the slice boundaries or for infinite if the infinite potential projection of each atom will be assigned to a single slice [MS21, REFERENCE/API/potentials/iam/Potential].

```
potential_unit = abtem.Potential(
    atoms = rotated_silicon,
    slice_thickness = t,
    sampling = s,
    parametrization = "labato"
    projection = "finite"
)
```

Code 5: Calculating the Potential Unit

To get the potential the "abtem.CrystalPotential()" function is used. It lowers the computational cost by using the previously calculated potential unit and repeating it a certain amount of times. This helps reducing the time needed to compute the potential for the different parameter sizes that have been previously defined [MS21, REFERENCE/API/potentials/iam/CrystalPotential].

```
potential = abtem.CrystalPotential(
    potential_unit = potential_unit,
    repetitions = (1, 1, 120)
)
```

Code 6: Calculating the Potential

With the now acquired potential the plane wave, representing the electron probe wave function, can be gained by the following two commands seen in code 7. The first function, taking energy as parameter, creates the plane wave and the following second function matches the grid from the plane wave to the potential. [MS21, REFERNCE/API/waves/PlaneWave],[MS21, REFERENCE/API/waves/BaseWaves].

```
plane_wave = abtem.PlaneWave(energy = 200e3)
plane_wave.grid.match(potential)
```

Code 7: Matching the Potential to the Plane Wave

The last step before computing the exit wave is the "MultisliceTransform()" function. A transformation that applies the multislice algorithm to the wave function. This is the part where the two multislice methods differ. The function takes the potential, the real or conventional method and if real space was selected a derivative accuracy (as seen in code 8) as parameters.

The derivative accuracy sets the accuracy of the finite difference derivative stencil [MS21, REFERENCE/API/multislice/MultisliceTransform].

```
transform_realspace = MultisliceTransform(
    potential,
    method = "realspace",
    derivative_accuracy = 8
)
transform_conventional = MultisliceTransform(
    potential,
    method = "conventional"
)
```

Code 8: Choosing the Multislice Transform method

Now the exit wave can be computed by building the Plane Wave and applying the Multislice Transform as seen in code 9. The whole code needed to compute the exit waves can be looked at in the Appendices at 15.

```
exit_wave = plane_wave.build().apply_transform(transform).compute()
```

Code 9: Computing the Exit Wave

2.4 Divergence Map

Using the realspace multislice method the calculation can diverge depending on the used slice thickness and sampling size. To be able to compare the same parameters as in the conventional method a divergence map was created by running the simulation for all possible combinations of slice thicknesses and sampling sizes previously defined in the parameters. Every time the calculation diverged the error was caught and saved. Allowing to later display a 2D map, where the calculation depending on the slice thickness and sampling would diverged. The method can be seen in code 10.

```
try:
    transform = MultisliceTransform(
        potential,
        method="realspace",
        derivative_accuracy = 8
    )
    exit_wave_rms = plane_wave.built().apply_transform(transform).compute()
except DivergedError:
    print("calculation diverged")
    divergence_map[i,j] = 1
else:
    exit_waves_rms.append(exit_wave_rms)
```

Code 10: Catching the Divergence Error

Visualizing the caught Divergence Errors allows to determine which slice thicknesses and sampling sizes can be later used to compare the two different methods. Looking at the Divergence

Map in Figure 8, for a sampling size 0.09 Å and higher all slice thicknesses can be used. A lower number in sampling means a better resolution and therefor better quality, by lowering the sampling size to 0.06 Å all but highest slice thickness can be used.

Figure 8: Divergence Map

2.5 Error Metric

One way to compare the two methods will be through the Error Metric 17 from [LR15, Eq.15].

$$\epsilon(z) = \sqrt{\frac{|\Psi(z) - \Psi_{MC}|^2}{|\Psi_{MC}|^2}} \tag{17}$$

Which will give the relative error $\epsilon(z)$ for the exit wave $\Psi(z)$ with respect to the most converged exit wave Ψ_{MC}. Where $z \in Z$ and Z is a set of all parameters used except the most converged one.

For different slice thicknesses the equation 17 can be written as seen in code 11. In this case the exit wave for the smallest slice thickness (last element in the list) is the most converged exit wave.

```
error_metric = []
for t in range(len(slice_thicknesses[1;])):
    num = np.linalg.norm(results[t].array - results[-1].array)**2
    den = np.linalg.norm(results[-1].array)**2
    error_metric.append(np.sqrt(num/den))
```

Code 11: Code for Error Metric

2.6 Intensity of Miller Indices

Another way of comparison is through the intensity of different miller indices. This can be achieved by using the "diffraction_patterns()" function in combination with the "index_diffraction_spots()" function. Enabling the visualization of Intensities for different miller indices as seen in Figure 9 and code 12.

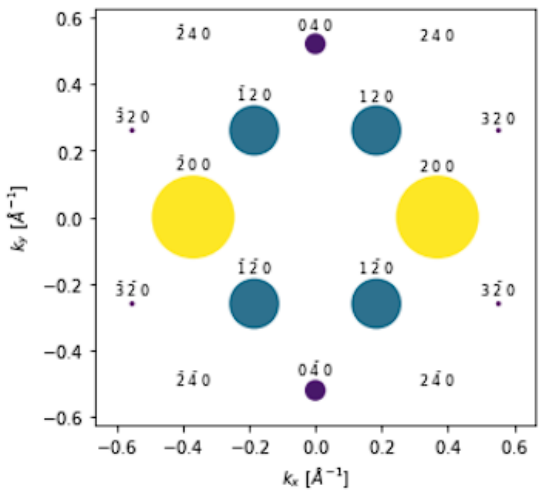


Figure 9: Visualization of Miller Indices

```
spots = results_diffraction_patterns().index_diffraction_spots(cell = rotated_silicon)
visu = spots.block_direct().crop(16).show(scale = 0.5)
visu.set_miller_index_annotations(threshhold = 0.001)
```

Code 12: Code for visualizing Miller Indices

The change of Intensity in respect to slice thickness, sampling size or energy can then be visualized via the following example code 14 for slice thickness. To safe space only the conventional method is presented.

3 Results

The aim of this Bachelor Thesis is to compare the Real Space and Conventional Multislice method in abTEM. This was done by using a set of different slice thicknesses, sampling sizes and energies, defined in section 2.2 and later redefined with the Introduction of the Real Space method and the Divergence Map in Section 2.4. As the Real Space method can diverge for large slice thicknesses and finer sampling sizes.

3.1 Slice Thickness

A sampling size of 0.06 Å was chosen, as a result of the Introduction of the Divergence map. Allowing only slice thicknesses ranging from $a/2$ to $a/256$ to be used. Leaving a set of 8 different slice thicknesses t. The energy was set to 200 keV.

$$t = \left\{ \begin{array}{l} 1.91979491, 0.95989746, 0.47994873, 0.23997436, \\ 0.11998718, 0.05999359, 0.0299968, 0.0149984 \end{array} \right\} \tag{18}$$

The exit waves were then calculated as already explained in the previous method section 2.

3.1.1 Error Metric

The Error Metric for different slice thicknesses, as seen in Figure 10, doesn't show quit a big difference between the two methods. Only at smaller slice thicknesses starting from $a/16$ to $a/128$ the relative error for the real space method can be seen to decrease faster then the conventional method. Meaning the real space method is for smaller slice thicknesses a little bit more accurate than the conventional method (The slice thickness $a/256$ was not included, because it is the most converged exit wave).

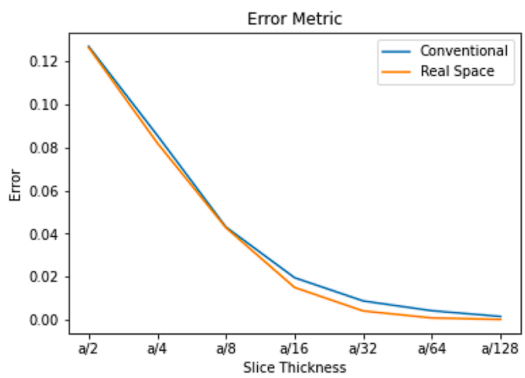


Figure 10: Slice Thickness - Error Metric

Examining the values directly from Table 4, as seen in Figure 10, starting from a slice thickness $a/16$ and smaller, a difference in quality between the two methods can be seen. But at these small slice thicknesses the Error of both methods is already very small $\sim 0.1\%$. By adding the time needed to compute the exit waves depending on their slice thickness to Figure 10 the relative error of the two methods can be compared with the time needed to compute each exit wave. Allowing a better judgment if the increase in quality over the calculation time is worth it. By looking at Figure 11 the time can be seen to be exponentially increasing for smaller slice thicknesses for both methods. As well as the real space method always taking longer to compute than the conventional method. Leading to a time difference of 31.86 seconds for $a/128$ between the conventional and real space method. Put in contrast with the slight increase in quality, where the error already is nearly negligible, the stark increase in calculation time does not justify the small decrease in error.

Slice Thickness [Å]	Conventional Error [%]	Real Space Error [%]	Conventional Time [s]	Real Space Time [s]
a/2	12.7057	12.6531	0.25	3.36
a/4	8.5721	8.2172	0.37	2.93
a/8	4.3093	4.2759	0.67	3.95
a/16	1.9541	1.5005	1.20	5.75
a/32	0.8736	0.4092	1.69	9.31
a/64	0.4193	0.0824	3.24	14.40
a/128	0.1531	0.0163	6.62	24.60

Table 4: Slice Thickness - Error Metric and Computational Time

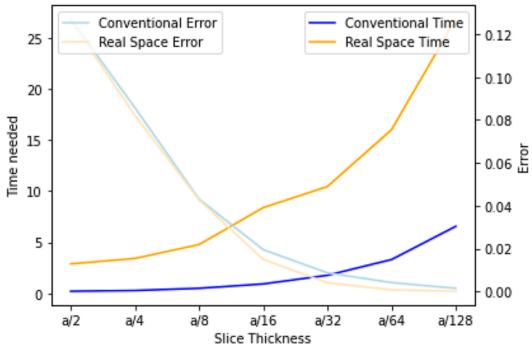


Figure 11: Slice Thickness - Error Metric and Calculation Time

3.1.2 Intensity of the Miller Indices

Eight different Miller Indices were selected and the change of intensity for the different slice thicknesses for both multislice methods were plotted. Viewing the different plots as seen in Figure 12, for all but the ($\bar{2}00$) miller index show for the real space method a slight decrease in intensity over the conventional method.

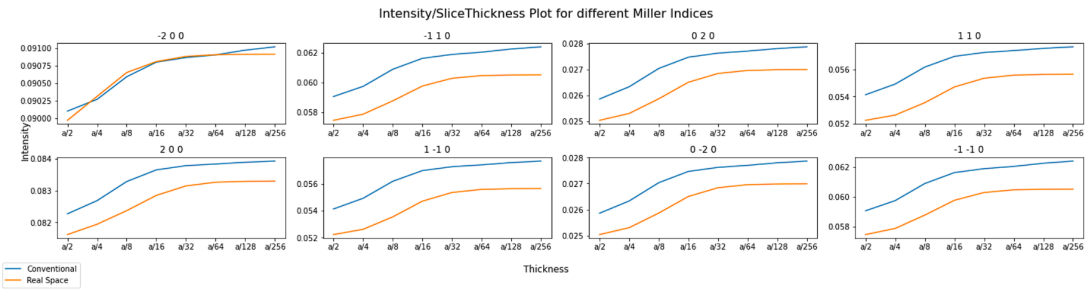


Figure 12: Intensity/Thickness Plots

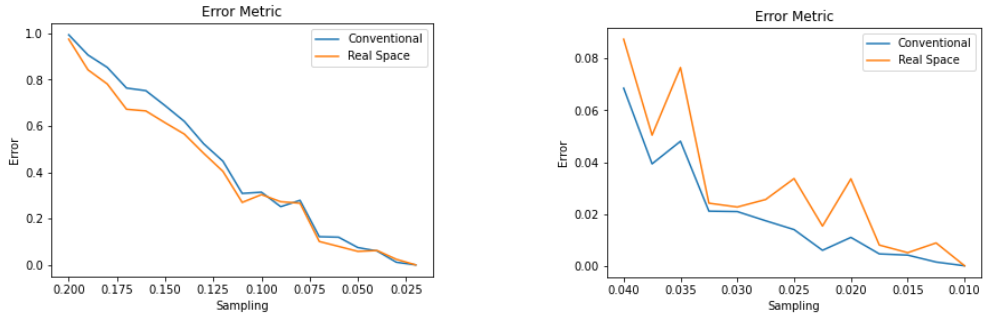
3.2 Sampling

Regarding the findings in Section 3.1, at a slice thickness of $a/64$ the intensity and the relativ error have converged and no meaningful change, except for an exponentially longer computational time, can be observed. For this reason the comparison of different sampling sizes for a fixed slice thickness of $a/64$ and an energy of 200 keV was chosen. Utilizing the Divergence Map from Section 2.4 all but the finest sampling size of 0.01 Å were computed.

$$s_1 = \left\{ \begin{array}{l} 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.10, \\ 0.11, 0.12, 0.13, 0.14, 0.15, 0.16, 0.17, 0.18, 0.19, 0.20 \end{array} \right\} \tag{19}$$

3.2.1 Error Metric

Similar to Figure 10 the relative error for the real space method is slightly lower then the conventional method. But inversely the difference in error decreases for finer sampling sizes as seen in Figure 13 and Table 5.



(a) (b)

Figure 13a shows the sampling - error metric between 0.20 Å and 0.02 Å with a slice thickness of a/64. Figure 13b shows the sampling - error metric between 0.040 Å and 0.010 Å with a slice thickness of a/256.

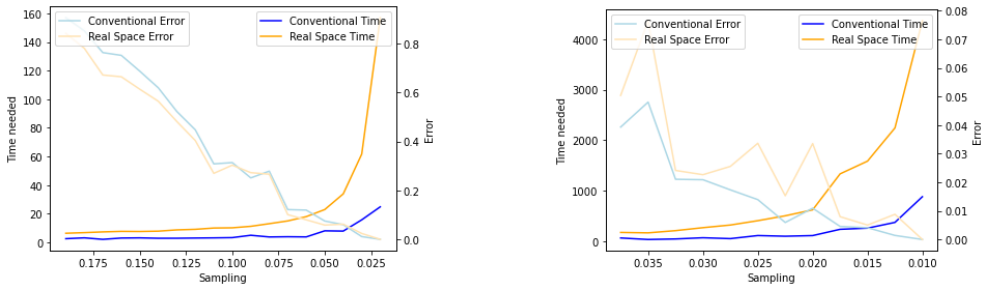
Figure 13: Sampling - Error Metric

The error metric for the different sampling sizes changes more drastically than that of the slice thickness, where the error only changes from $\sim$ 12% to $\sim$ 0.15%. The relative error for the chosen set of sampling sizes 19 changes from $\sim$ 99% to $\sim$ 1%. This is an result of the high change in quality over the different sampling sizes. By disregarding the poorer sampling sizes and only focusing on sample sizes ranging from 0.0400 Å to 0.0100 Å as seen in set 20, a convergence can be observed. As to the Divergence Map 8, trying to compute such fine sampling sizes require a much thinner slice thickness of $a/256$ instead of the previously used $a/64$ for Figure 13a.

$$s_2 = \left\{ \begin{array}{l} 0.0100, 0.0125, 0.0150, 0.0175, 0.0200, 0.0225, 0.0250, \\ 0.0275, 0.0300, 0.0325, 0.0350, 0.0375, 0.0400 \end{array} \right\} \tag{20}$$

Sampling [Å]	Conventional Error [%]	Real Space Error [%]	Conventional Time [s]	Real Space Time [s]
0.02	0.0000	0.0000	24.7897	157.0272
0.03	1.1317	2.4071	15.7082	61.6822
0.04	6.0017	6.2009	7.8409	33.8530
0.05	7.4885	5.8363	8.0242	22.9050
0.06	11.9596	7.9837	3.7752	17.9506
0.07	12.1686	10.1054	3.8909	14.9430
0.08	27.8753	26.6343	3.7358	13.0204
0.09	25.1219	27.2696	4.9173	11.1667
0.10	31.3780	30.3355	3.2715	10.1319
0.11	30.8508	26.9740	3.0950	9.9728
0.12	44.7768	40.3965	2.9775	9.0675
0.13	52.2887	48.1875	2.8344	8.6710
0.14	61.9437	56.4540	2.8189	7.7870
0.15	68.7403	61.3646	3.0946	7.5569
0.16	75.2150	66.4627	2.9763	7.6318
0.17	76.3379	67.1797	2.0589	7.2682
0.18	85.2468	78.0957	3.1108	6.7580
0.19	90.6039	84.1907	2.5253	6.3224
0.20	99.3245	97.4328	2.3094	6.7182

Table 5: Sampling - Error Metric and Computational Time (1)



(a) (b)

Figure 14a shows the sampling - error metric - time between 0.20 Å and 0.02 Å with a slice thickness of a/64.
Figure 14b shows the sampling - error metric - time between 0.040 Å and 0.010 Å with a slice thickness of a/256.

Figure 14: Sampling - Error Metric and Computational Time

Adding the time needed to compute the exit waves allows a better interpretation of the relation of the decreasing error to the increasing computational time. As seen in Figure 14, the time only starts to increase noticeably at around 0.1 Å and then starts to increase exponentially, with the real space method taking an unreasonably amount of time more then the conventional method for very fine sampling sizes. For different sampling sizes the change in error between the two methods changes over the span of chosen sampling sizes. For low quality sampling sizes as can be seen in Figure 14a the error metric for the real space multislice method seems to be of better quality. But by changing the slice thickness from $a/64$ to $a/256$ and looking at a finer sampling size set s_2 20 as can be seen in Figure 14b the conventional method has surpassed the real space method in accuracy. This is interesting as Figure 10 shows an increase in accuracy for the real space method. Meaning a change in sampling has a bugger impact than the slice thickness.
The data for Figure 13 and 14 can be looked at Table 6 and 5.

Sampling [Å]	Conventional Error [%]	Real Space Error [%]	Conventional Time [s]	Real Space Time [s]
0.0100	0.0000	0.0000	881.8807	4364.5319
0.0125	0.1426	0.8807	373.7935	2243.1145
0.0150	0.4110	0.5047	259.7291	1586.3640
0.0175	0.0458	0.7985	236.8181	1334.2248
0.0200	1.0948	3.3562	112.9981	622.4619
0.0225	0.5959	1.5308	100.1337	504.4552
0.0250	1.3952	3.3663	113.6507	407.6493
0.0275	1.7360	2.5580	52.6729	321.8172
0.0300	2.0910	2.2666	69.9835	269.3447
0.0325	2.1070	2.4159	46.1025	210.5294
0.0350	4.8062	7.6527	35.3519	166.8074
0.0375	3.9329	5.0417	65.6240	175.1669
0.0400	6.8546	8.7447	37.9214	179.5979

Table 6: Slice Thickness - Error Metric and Computational Time (2)

3.2.2 Intensity of the Miller Indices

Viewing the change of Intensity of the Miller indices for different sampling sizes, as seen in Figure 15, shows no significant difference between the two methods.

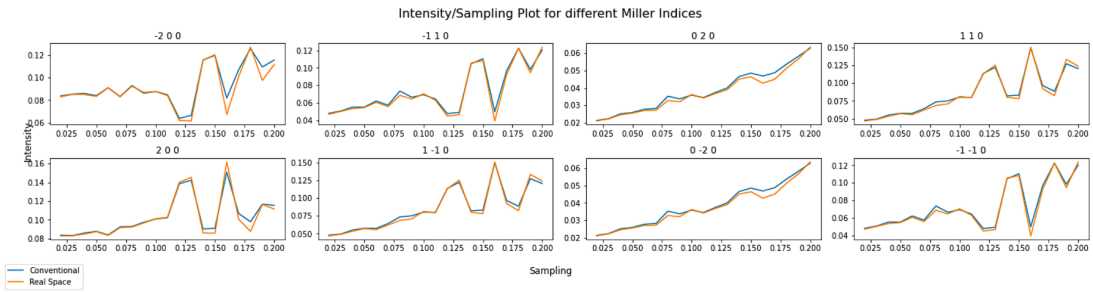


Figure 15: Sampling - Intensity of different Miller Indices

3.3 Energy

The real space method does divergence for large slice thicknesses and fine sampling sizes. A change in energy should not result in a divergence of the real space method. Therefor no necessary limitation for the energy parameters are needed.

$$e = \left\{ \begin{array}{l} 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, \\ 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300 \end{array} \right\} keV \tag{21}$$

As in section 3.2 a slice thickness of $a/64$ was chosen. For the sampling size the error does not converge for the parameters used in 3.2. To keep computational time low a compromise in accuracy was made and a sampling size of 0.06 Å was chosen.

3.3.1 Error Metric

Viewing Figure 16 no significant difference between the two methods can be observed.

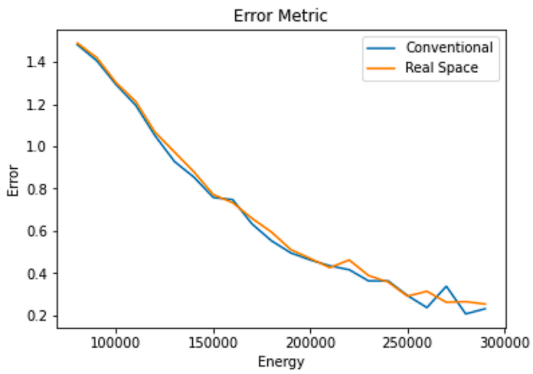


Figure 16: Energy - Error Metric

A notable change in computational time for different energy levels can not be observed. But because the real space method already takes longer to compute as seen in the slice thickness and sampling size series. It as well takes longer to compute in the energy series as seen in Figure 17.

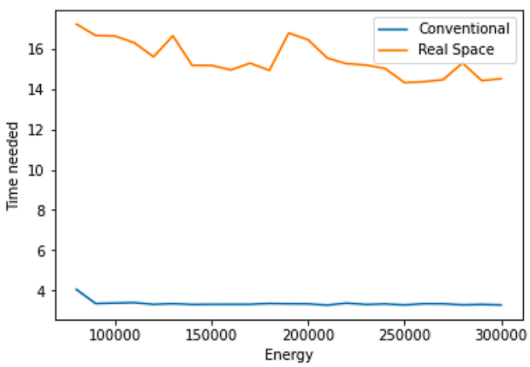


Figure 17: Energy - Time needed to compute different energy levels

Sampling [Å]	Conventional Error [%]	Real Space Error [%]	Conventional Time [s]	Real Space Time [s]
80 000	148.2067	148.9313	4.8693	17.9626
90 000	140.6011	142.0495	3.4537	16.5577
100 000	129.2122	130.1207	3.3950	16.1978
110 000	119.5925	121.3415	3.4780	16.1652
120 000	105.0918	106.7376	3.3360	14.7717
130 000	92.8894	97.3726	3.3340	15.6380
140 000	85.4754	87.9819	3.3270	14.8000
150 000	75.8410	77.1754	3.2700	14.6895
160 000	74.7960	73.3514	3.2724	14.7628
170 000	63.2466	65.8870	3.3020	14.6066
180 000	55.2965	59.4252	3.2990	15.5882
190 000	49.5468	51.0600	3.2840	15.0690
200 000	46.2183	46.9066	3.3070	14.8165
210 000	43.4497	42.5918	3.2900	14.6933
220 000	41.6094	46.2173	3.2970	14.7172
230 000	36.3075	38.8196	3.6370	14.6819
240 000	36.3283	35.7900	3.3116	16.1567
250 000	29.3729	29.0975	3.3210	14.6610
260 000	23.6793	31.3831	3.3380	13.8810
270 000	33.7761	26.1927	3.3080	14.1929
280 000	20.7054	26.4773	3.2980	14.2718
290 000	23.1258	25.3415	3.2990	15.3816
300 000	0.0000	0.0000	3.3020	15.9360

Table 7: Energy - Error Metric and Computational Time

3.3.2 Intensity of the Miller Indices

No difference between the methods can viewed by looking at Figure 18.

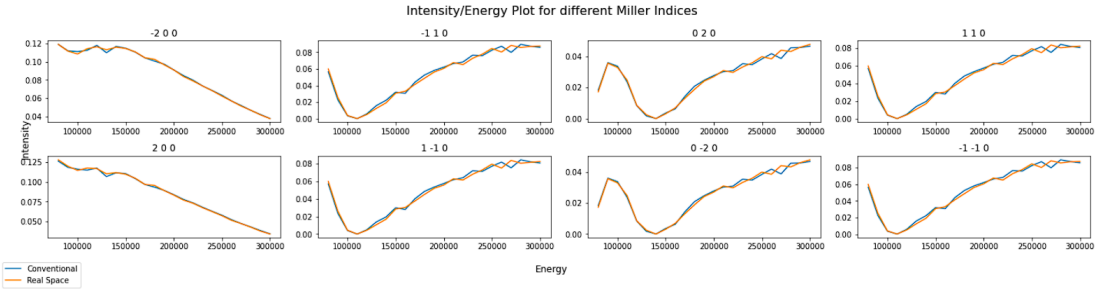


Figure 18: Energy - Intensity of different Miller Indices

4 Discussion & Conclusion

The aim of this bachelor thesis was to compare the real space and conventional method in abtem and to find out if the real space method has any advantages over the conventional method, especially in regards to computational time and accuracy.

As seen in Section 3.1 the real space method has some minor advantages in accuracy for smaller slice thicknesses but the exponential increase in computational time does not justify its use.
This trend of increased computational cost for minimal decrease of error does even reverse when looking at the sampling-error curve from Figure 14 in section 3.2. Here the error difference between the two methods decreases for finer sampling sizes.
The results from section 3.3 do not show any error difference between the two methods in relation to energy, except for as already stated for the other two parameters an increase in computational time.

The gathered data from Section 3, for the used parameters defined in Section 2.2, do not confirm the previously anticipated advantages from the real space multislice method used in abTEM. The underlying problem lies in the stark difference between the computational cost and the minimal, if even, decrease in error, compared to the conventional multislice method.
To a similar conclusion came the Article "Multislice algorithms revisited: Solving the Schrödinger equation numerically for imaging with electrons" by C. Wacker and R.R. Schröder [WS15]. Who as well concluded that their real space algorithm were not sufficient to beat the conventional multislice method and were not affected by convergence problems.

In conclusion the conventional- and real space multislice method for different parameters were computed in abTEM and their results with respect to computational time and accuracy were compared. An anticipated higher performing real space method could not be observed. Leaving the conventional multislice method as the preferred one.

5 Appendices

```
#Create 2D Map
map = plt.pcolormesh(
divergence_map,
edgecolors = "k",
linewidth = 2,
cmap = "binary"
)

#Define Positions of y/x- ticks
xticks = plt.xticks()[0][:-1] + 0.5
yticks = np.arange(start = 0, stop = 20, step = 1) + 0.5

#Change x/y - ticks position and label
plt.xticks(
ticks = xticks,
labels = [f"a/{n}" for n in thickness_fractions],
rotation = 20
)
plt.yticks(
ticks = yticks,
labels = np.round(sampling, decimals = 2)
)

#Define Legend
patches = [
mpatches.Patch(color = (0.0, 0.0, 0.0, 1.0), label = "Diverged"),
mpatches.Patch(color = (1.0, 1.0, 1.0, 1.0), label = "Converged")
]

#Plotting
plt.legend(
handles=patches,
loc = "center right",
bbox_to_anchor = (1.3, 0.5, 0.0, 0.0),
borderaxespad = 0
)
plt.title("Divergence Map")
plt.xlabel("Slice Thickness")
plt.ylabel("Sampling")
plt.show()
```

Code 13: Code for visualizing the Divergence Map

```
df = spots[:,0].to_dataframe()
df.index = slice_thicknesses[1:]

fig, ((
ax1, ax2, ax3, ax4),
(ax5,ax6,ax7,ax8)
) = plt.subplots(2,4 figsize=(20,5))
fig.supxlabel("Thickness")
fig.supylabel("Intensity")
fig.suptitel(
"Intensity/SliceThickness Plot for different Miller Indices",
frontsize=16
)

miller_indice = [
"-2 0 0",
"-1 1 0",
"0 2 0",
"1 1 0",
"2 0 0",
"1 -1 0",
"0 -2 0",
"-1 -1 0"
]
ax_list = fig.axes

for ax in ax_list:
miller_indexx = miller_indice[ax_list.index(ax)]
ax.plot(
[f"a/{n}" for n in thickness_fractions[1:]],
np.array(df[miller_index]),
label "Conventional",
)
ax.set_title(miller_index)

handles, labels = ax.get_legend_handles_labels()
fig.legend(
handles,
labels,
loc="lower left",
bbox_to_anchor = (0.0, -0.05, 0.0, 0.0)
)

plt.tight_layout()
plt.show()
```

Code 14: Code for visualizing Miller Indices

```python
exit_waves = []
times = []

for t in slice_thicknesses[1:]:
    t_0 = time.time()
    potential_unit = abtem.Potential(
    atoms = rotated_silicon,
    slice_thickness = t,
    sampling = 0.06,
    projection = "finite",
)

potential = abtem.CrystalPotential(
    potential_unit = potential_unit,
    repetitions = (1, 1, 120),
)

plane_wave = abtem.PlaneWave(energy = 200e3)
plane_wave.grid.match(potential)

try:
    transform = MultisliceTransform(
        potential,
        method="realspace",
        derivative_accuracy=8
    )
    exit_wave = plane_wave.build().apply_transform(transform_rms).compute()
    except DivergedError:
    print("calulation diverged")
else:
    exit_waves.append(exit_wave)
    times.append(time.time() - t_0)
```

Code 15: Full Multislice Code in abTEM

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